

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-	Team No-	Project Title-
Team Lead Roll Number		2520030430
Team Lead Name		KOTHA GOLLA PRAGNYA
Team Member Roll Number		2520030482
Team Member Name		MATTI AVISHYA

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The **Anti-Ragging Complaint Portal** is designed to provide students with a safe, anonymous, and structured way to report incidents of ragging within a college environment. Built using modern web technologies such as React.js, Node.js, Express.js, and MySQL, the portal ensures seamless complaint submission, secure data handling, and efficient case management. Students can lodge complaints anonymously, track the progress of their cases, and receive updates through notifications. The system also integrates committee assignment features, enabling administrators to delegate cases to appropriate disciplinary committees for review and resolution. With a responsive design powered by Bootstrap or Tailwind CSS, the portal is accessible across devices, ensuring ease of use for both students and administrators.

Beyond complaint handling, the portal emphasizes accountability and transparency through modules like case history, committee notes, and statutory report generation. Administrators can generate detailed reports for compliance with anti-ragging regulations, export case data for audits, and maintain a centralized record of all complaints. The admin panel provides oversight with tools for monitoring complaint trends, managing committee assignments, and ensuring timely resolution. By combining anonymity, structured workflows, and robust reporting, the portal not only empowers students to speak up without fear but also strengthens institutional mechanisms to uphold discipline and safety on campus.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Complaint Form Anonymity Option Case Assignment Complaint History Notifications	2520030430	
Status Tracking Export Data Committee Notes Report Generation Admin Panel	2520030482	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

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- **Complaint Form & Anonymity Option:** Builds the front-end form with React.js, applies declarative UI and Virtual DOM concepts, and ensures secure anonymous submission using Node.js/Express.
- **Case Assignment:** Implements committee assignment logic, integrates REST APIs with Axios, and applies immutability principles for secure data handling.
- **Complaint History:** Designs searchable archives with pagination, applies functional programming for filtering, and ensures efficient rendering pipeline.
- **Notifications:** Creates real-time notification system using async/await and event-driven programming, ensuring reactive state updates.

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- **Status Tracking:** Develops student dashboard with React Router and Redux, applies unidirectional data flow and reconciliation for efficient updates.
- **Committee Notes:** Implements secure role-based access for committee members, applies component-driven design and immutability for note storage.
- **Report Generation:** Automates statutory report creation in PDF/CSV formats, applies promise chains and error boundaries for robust UI updates.
- **Export Data:** Provides export functionality with module patterns, ensures correctness using TypeScript interfaces.
- **Admin Panel:** Designs responsive admin interface with Bootstrap/Tailwind CSS, applies architectural concepts like composition and reactive state management.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Front-End: HTML5, CSS3, JavaScript (ES6+), React.js, React Router, Context API/Redux, Bootstrap/Tailwind CSS

Back-End: Node.js, Express.js, REST APIs, Axios

Database: MySQL (optional, for complaint storage and tracking)

Testing & Tools: Postman (API testing), npm (package management), LocalStorage (temporary client-side storage)

Guide Name and Signature (remarks if any) _____